OpenFOAM Assignment 3

COE 347: Intro to Computational Fluid Dynamics

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4 Governing Equations

4.1 Euler Equations in Vector Form

The governing equations are the two-dimensional Euler equations for a compressible ideal gas with ratio of specific heats $\gamma = 1.4$. In conservation form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = \mathbf{0} \tag{1}$$

The state vector $\mathbf{U}$, and the x- and y-flux vectors $\mathbf{F}$ and $\mathbf{G}$ are:

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \qquad \mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ \rho u H \end{bmatrix}, \qquad \mathbf{G} = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ \rho v H \end{bmatrix} \tag{2}$$

where $H = E + p/\rho$ is the specific total enthalpy and E is the specific total energy:

$$E = e + \tfrac{1}{2}\big(u^2 + v^2\big) \tag{3}$$

The system is closed by the equation of state for a calorically perfect ideal gas:

$$p = (\gamma - 1)\big[\rho E - \tfrac{1}{2}\rho\big(u^2 + v^2\big)\big] \tag{4}$$

with internal energy $e = p/[\rho(\gamma - 1)]$ and temperature $T = p/(\rho R)$, where R is the specific gas constant.

4.2 Definitions: Local Mach Number and Sound Speed

The local isentropic sound speed a and the local Mach number M are defined as:

$$a = \sqrt{\frac{\gamma\, p}{\rho}}, \qquad M = \frac{|\mathbf{u}|}{a} = \frac{\sqrt{u^2 + v^2}}{a} \tag{5}$$

These definitions are used directly when post-processing simulation output in ParaView to compute the Mach number and sound speed fields.

4.3 Analytical Estimate — Stagnation Pressure at the Step

The supersonic Mach 3 flow impinges on the vertical face of the forward-facing step at $(x, y) = (0.6, 0)$, producing a normal shock. The stagnation pressure at that point is estimated by (1) applying the Rankine–Hugoniot normal shock jump relations to find the post-shock state, then (2) isentropically decelerating the subsonic post-shock flow to stagnation.

Step 1 — Normal shock jump relations ($M_1 = 3$, $p_1 = 1$ Pa, $\gamma = 1.4$):

$$\frac{p_2}{p_1} = \frac{2\gamma M_1^2 - (\gamma - 1)}{\gamma + 1} = \frac{2(1.4)(9) - 0.4}{2.4} = \frac{31.2}{3.0} \approx 10.333 \tag{6}$$

$$p_2 = 10.333 \text{ Pa} \tag{7}$$

$$M_2^2 = \frac{(\gamma - 1)M_1^2 + 2}{2\gamma M_1^2 - (\gamma - 1)} = \frac{(0.4)(9) + 2}{(2.8)(9) - 0.4} = \frac{5.6}{24.8} = 0.2258 \tag{8}$$

$$M_2 = 0.4752 \quad \text{(subsonic, as expected)} \tag{9}$$

Step 2 — Isentropic deceleration from state 2 to stagnation ($M_2 = 0.4752$, $\gamma = 1.4$):

$$\frac{p_0}{p_2} = \left(1 + \frac{\gamma - 1}{2} M_2^2\right)^{\gamma/(\gamma-1)} = (1 + 0.2 \times 0.2258)^{3.5} = (1.04516)^{3.5} \approx 1.1672 \tag{10}$$

$$p_0 = 1.1672 \times 10.333 \approx 12.06 \text{ Pa} \tag{11}$$

The analytical stagnation pressure at the foot of the step for $M_1 = 3$, $p_1 = 1$ Pa is therefore:

$$\boxed{p_0 \approx 12.06 \text{ Pa}} \tag{12}$$

For reference, the isentropic (no-shock) free-stream total pressure would be $p_{0,\text{isentropic}} = p_1\left(1 + \frac{\gamma-1}{2} M_1^2\right)^{\gamma/(\gamma-1)} = 36.73$ Pa. The normal shock therefore reduces the stagnation pressure by a factor of 0.328, consistent with significant total pressure loss across a strong shock.

5 First Solution on the Coarse Mesh

5.1 Case Setup and Boundary Conditions

The case is set up from the class repository under `OpenFOAM/current/mach-3-wind-tunnel-with-step`, run with OpenFOAM v7 using the solver `rhoCentralFoam`. The numerical flux scheme is the Kurganov–Tadmor central-upwind scheme (`fvSchemes: fluxScheme Kurganov`), with van Leer reconstruction for ρ, $\mathbf{U}$, and T, and first-order Euler time integration.

The gas is modeled as an inviscid ($\mu = 0$), calorically perfect gas with $\gamma = 1.4$. The thermophysical properties are normalized so that the sound speed is exactly $a = 1$ m/s at $T = 1$ K: molecular weight $\mathcal{M} = 11640.3$ g/mol giving $R = 8314/11640.3 = 0.7143$ J/(kg K), $C_p =$

2.5 J/(kg K), $C_v = C_p/\gamma = 1.786$ J/(kg K). No turbulence model is used (`simulationType laminar`).

Table 1 summarizes the boundary conditions as configured in the `0/` directory.

Table 1: Boundary conditions (from `0/p`, `0/T`, `0/U`).

Patch	Type	p	T	U	Notes
`inlet`	patch	`fixedValue = 1 Pa`	`fixedValue = 1 K`	`fixedValue = (3 0 0) m/s`	$M = 3$ supersonic inflow
`outlet`	patch	`zeroGradient`	`inletOutlet`	`inletOutlet`	Supersonic outflow
`top`	symmetryPlane	`symmetryPlane`	`symmetryPlane`	`symmetryPlane`	Top tunnel wall
`bottom`	symmetryPlane	`symmetryPlane`	`symmetryPlane`	`symmetryPlane`	Inlet floor ($x < 0.6$)
`obstacle`	patch	`zeroGradient`	`zeroGradient`	`slip`	Step faces (inviscid wall)
`defaultFaces`	empty	`empty`	`empty`	`empty`	2D front/back faces

With $\mathbf{U} = (3,\ 0,\ 0)$ m/s and $a = \sqrt{\gamma p/\rho} = \sqrt{1.4 \times 1/1.4} = 1$ m/s at the inlet, the inlet Mach number is $M = 3/1 = 3$, consistent with the problem specification. The internal field is initialized uniformly to the inlet conditions throughout the domain.

5.2 Coarse Mesh and Time Step

The coarse mesh is defined in `blockMeshDict` using three hexahedral blocks that together tile the L-shaped domain (step region excluded). The geometry spans $x \in [0,\ 3]$ m, $y \in [0,\ 1]$ m, with the step occupying $x \in [0.6,\ 3]$ m, $y \in [0,\ 0.2]$ m. The domain is one cell thick ($z \in [-0.05,\ 0.05]$ m) with empty boundary conditions front and back for the 2D treatment.

Block 1 covers the inlet floor region ($x \in [0,\ 0.6]$ m, $y \in [0,\ 0.2]$ m) with $24 \times 8 = 192$ cells. Block 2 covers the upper inlet region ($x \in [0,\ 0.6]$ m, $y \in [0.2,\ 1]$ m) with $24 \times 32 = 768$ cells. Block 3 covers the main channel downstream of the step ($x \in [0.6,\ 3]$ m, $y \in [0.2,\ 1]$ m) with $96 \times 32 = 3{,}072$ cells. All cells are uniform with $\Delta x = \Delta y = 0.025$ m. The total coarse mesh contains 4,032 cells.

Time integration uses the first-order explicit Euler scheme. The time step is adjusted automatically (`adjustTimeStep yes`) to enforce a maximum Courant number of $Co_{\max} = 0.2$ (`controlDict: maxCo 0.2`). With a maximum wave speed of $|\mathbf{U}| + a = 4$ m/s and $\Delta x = 0.025$ m, this gives

$$\Delta t \approx \frac{Co_{\max}\,\Delta x}{|\mathbf{U}| + a} = \frac{0.2 \times 0.025}{4} = 0.00125 \text{ s.}$$

The simulation is advanced to $T = 4$ s (`endTime 4.0` in `controlDict`), writing output every 0.1 s.

5.3 ParaView Figures at $T = 4$ s

The following eight figures show the converged flow field at $T = 4$ s on the coarse mesh.

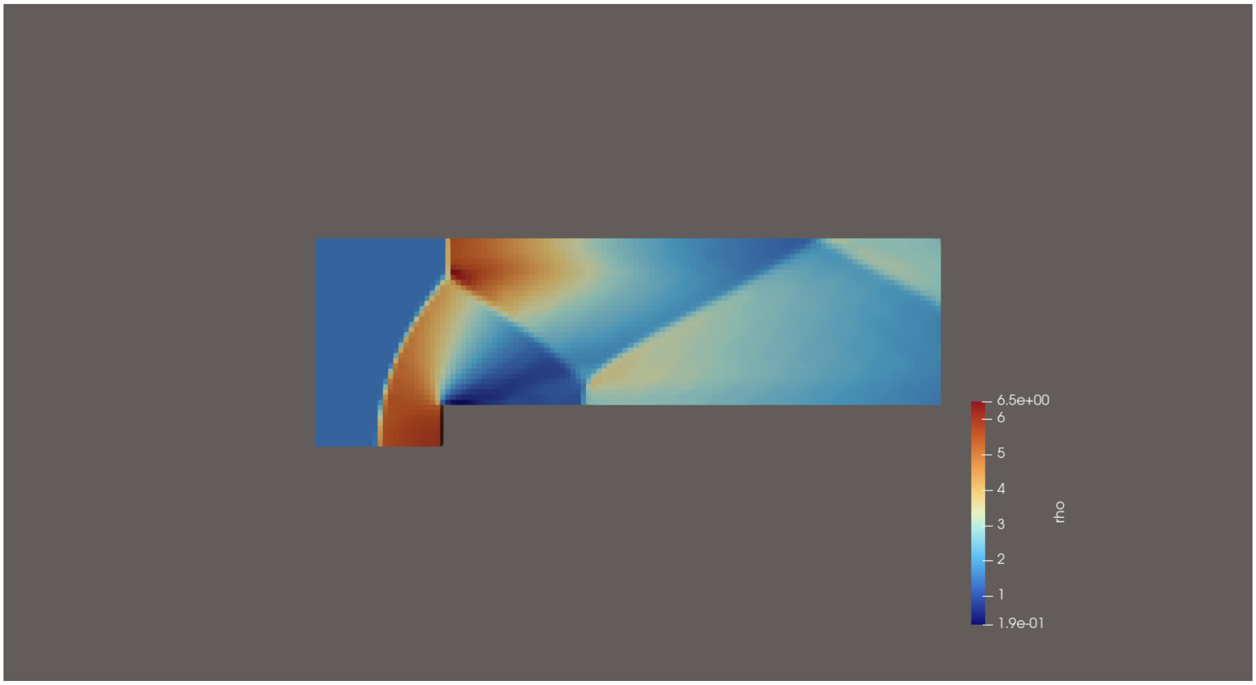

Figure 1: Density ρ [kg/m^3] at $T = 4$ s — coarse mesh.

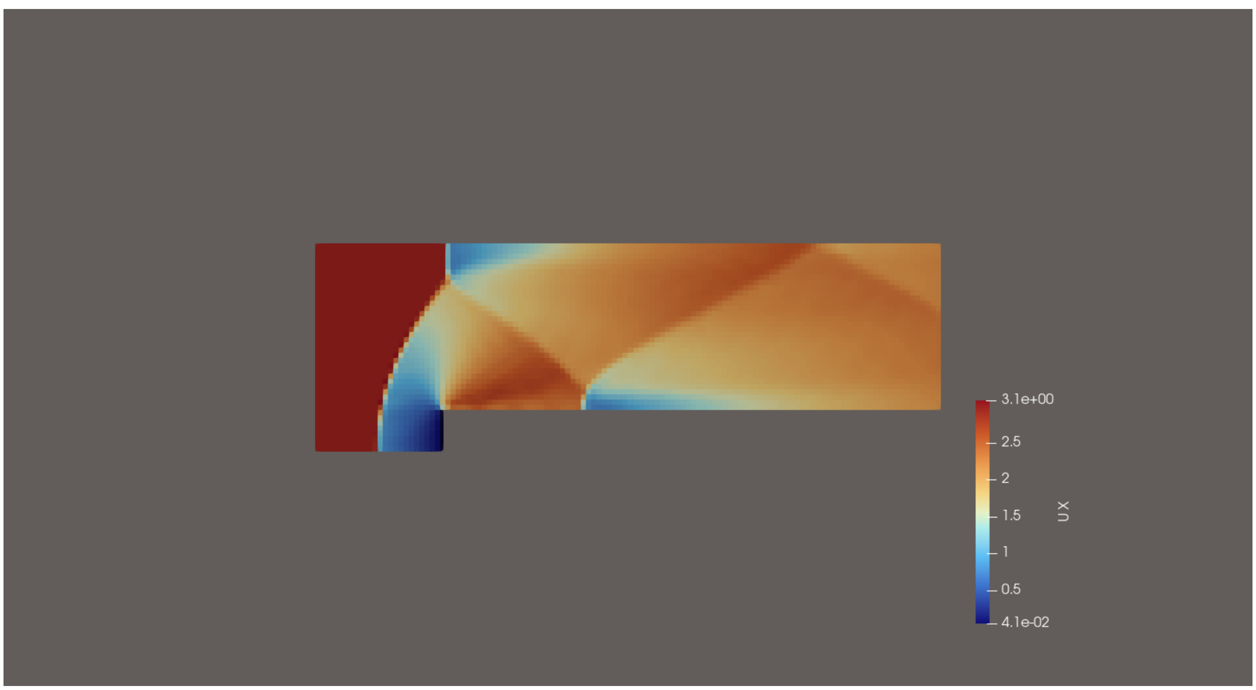

Figure 2: x-velocity component u [m/s] at $T = 4$ s — coarse mesh.

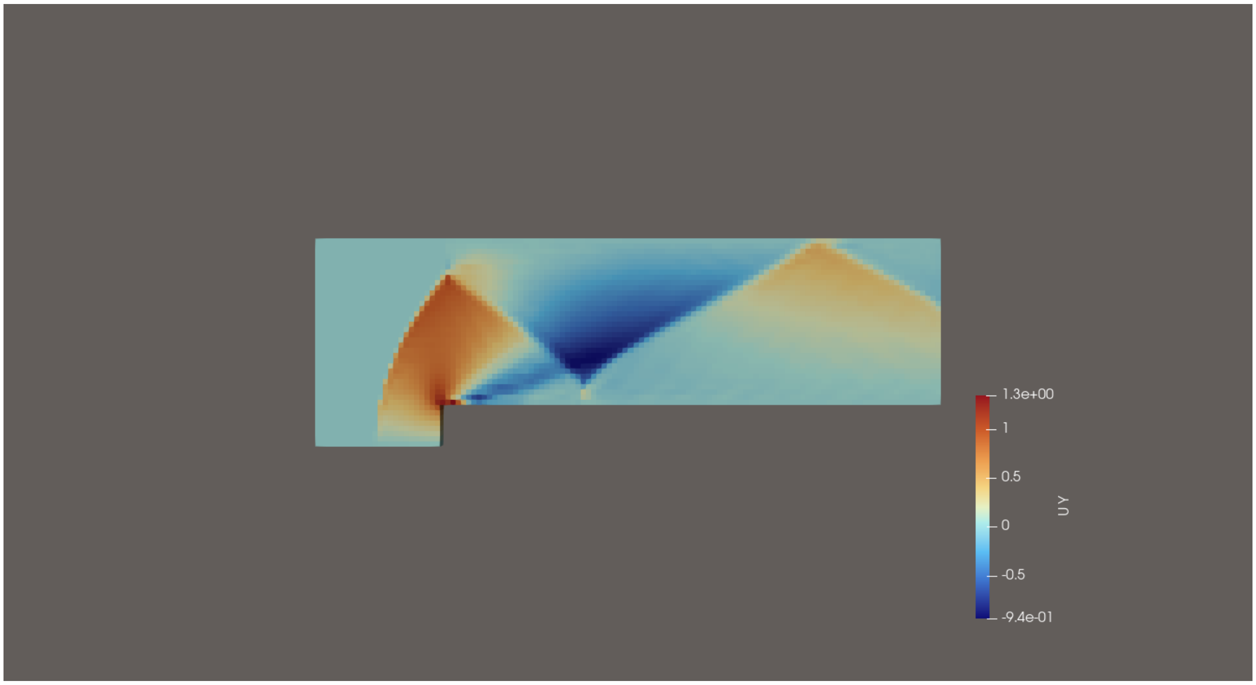

Figure 3: y-velocity component v [m/s] at $T = 4$ s — coarse mesh.

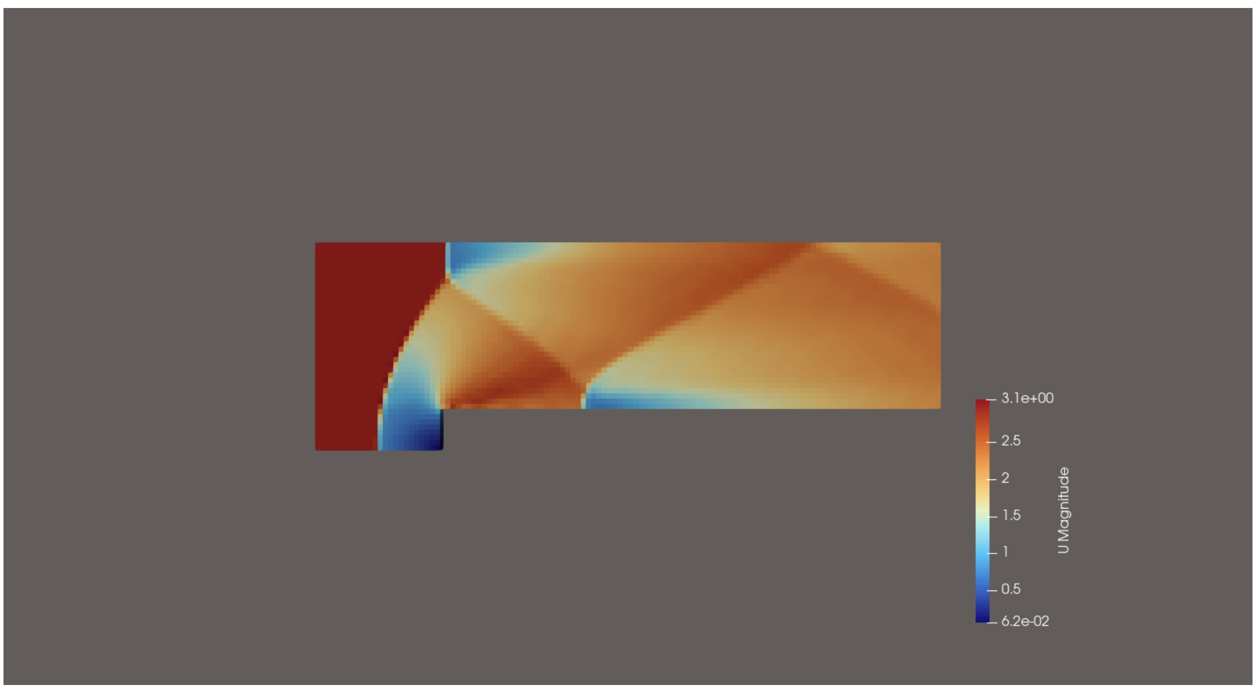

Figure 4: Velocity magnitude $|\mathbf{u}|$ [m/s] at $T = 4$ s — coarse mesh.

Figure 5: Pressure p [Pa] at $T = 4$ s — coarse mesh.

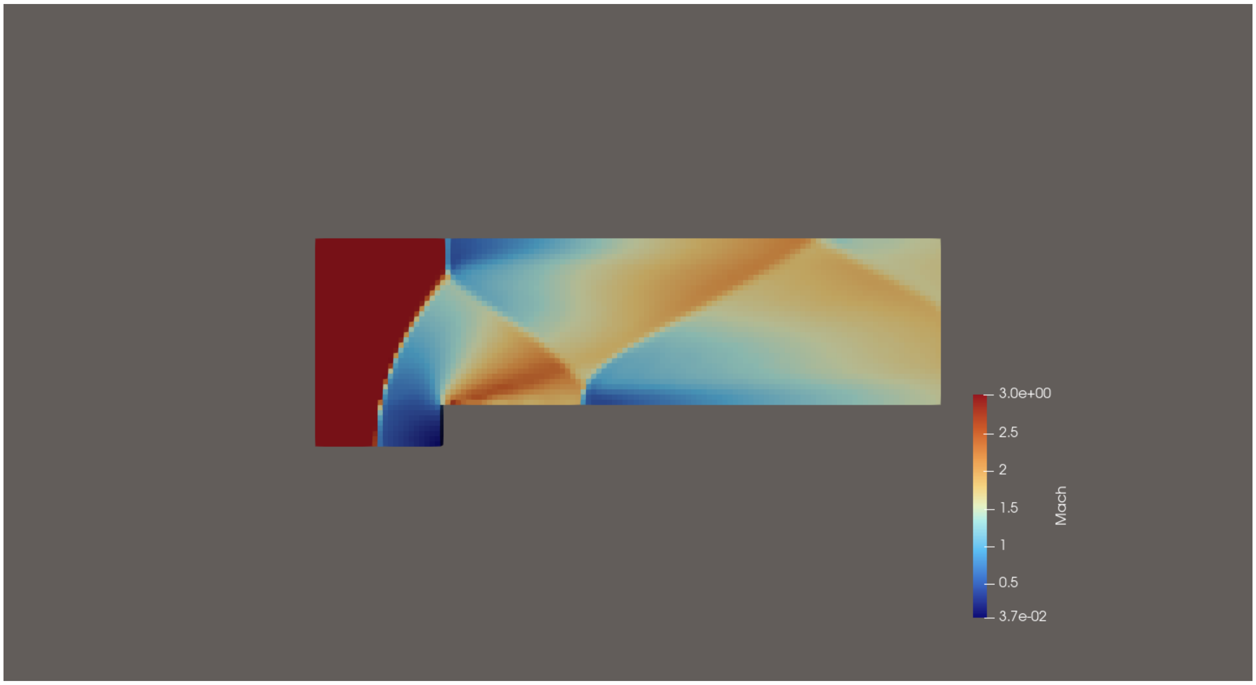

Figure 6: Mach number $M = |\mathbf{u}|/a$ at $T = 4$ s — coarse mesh.

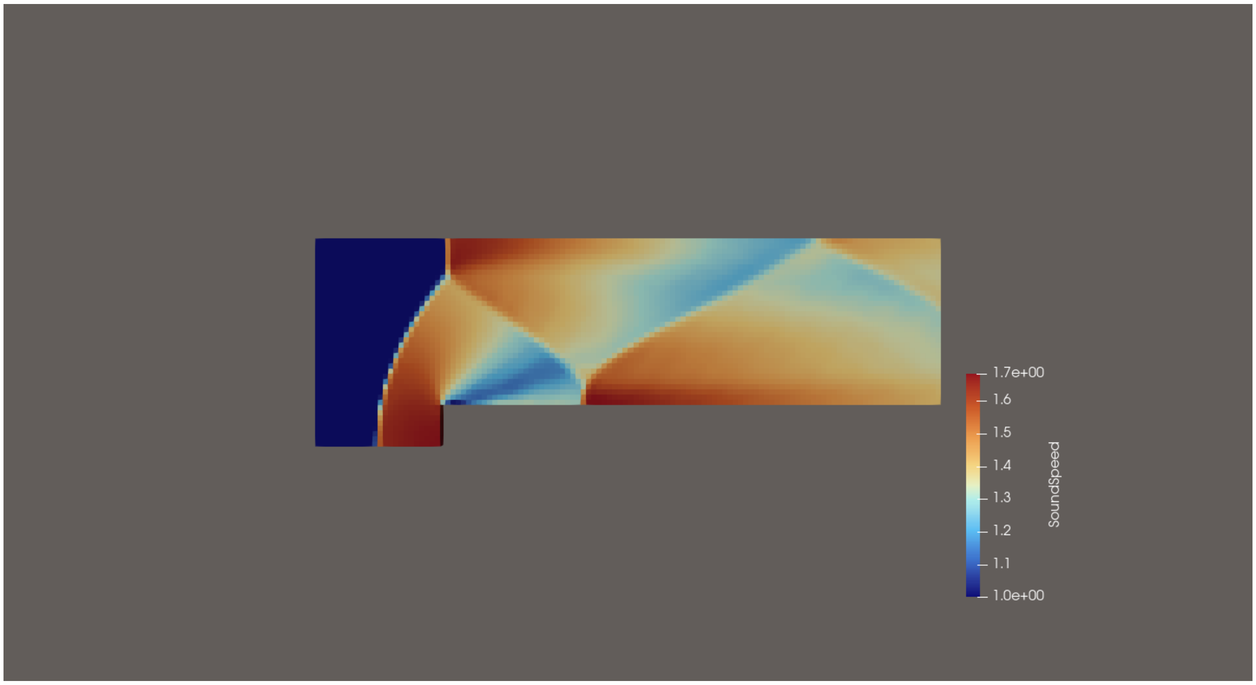

Figure 7: Sound speed a [m/s] at $T = 4$ s — coarse mesh.

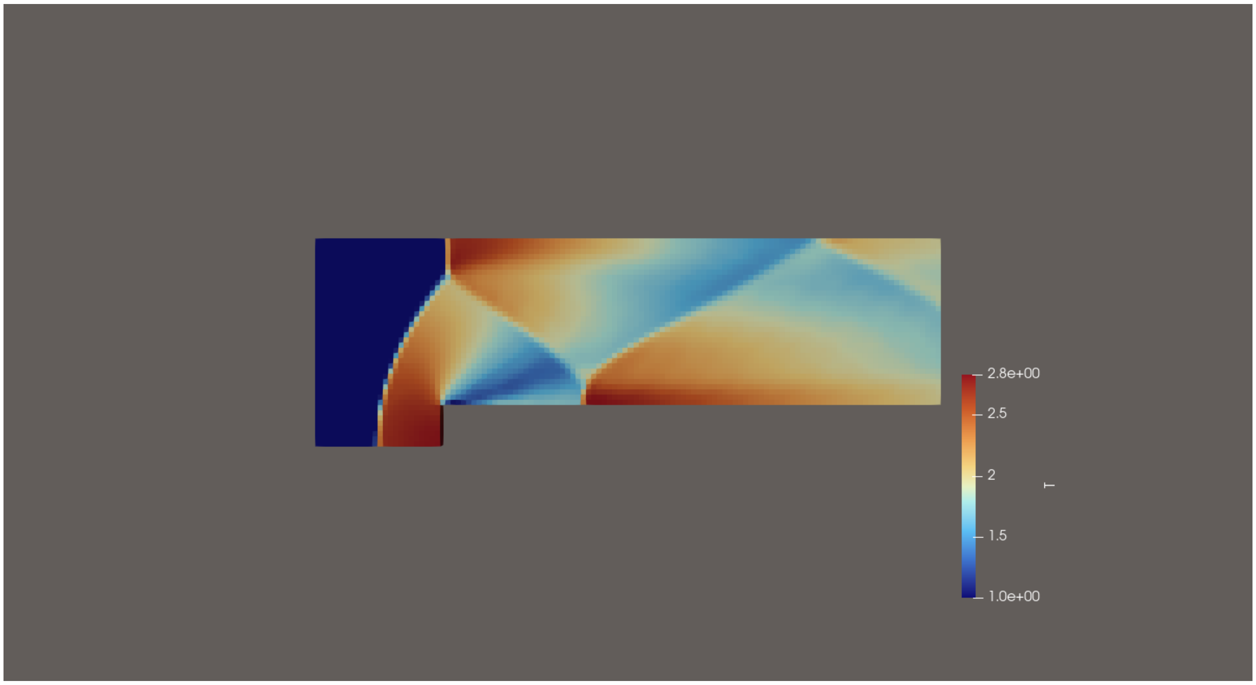

Figure 8: Temperature T [K] at $T = 4$ s — coarse mesh.

6 Convergence of the Solution

6.1 Mesh Details

Table 2 summarizes the mesh configurations used in the convergence study. For each mesh the time step was chosen to satisfy the CFL stability condition.

Table 2: Mesh parameters used in the convergence study.

Mesh Level	Cells (x)	Cells (y)	Total Cells	Δt (s)	Notes
Coarse	24 / 96	8 / 32	4,032	$\sim$0.00125	Base mesh
Medium	48 / 192	16 / 64	16,128	smaller	2× refinement
Fine	96 / 384	32 / 128	64,512	smaller	4× refinement

6.2 Effect of Spatial Resolution — ParaView Figures

Figures below show the effect of mesh refinement on the pressure and density fields at $T = 4$ s.

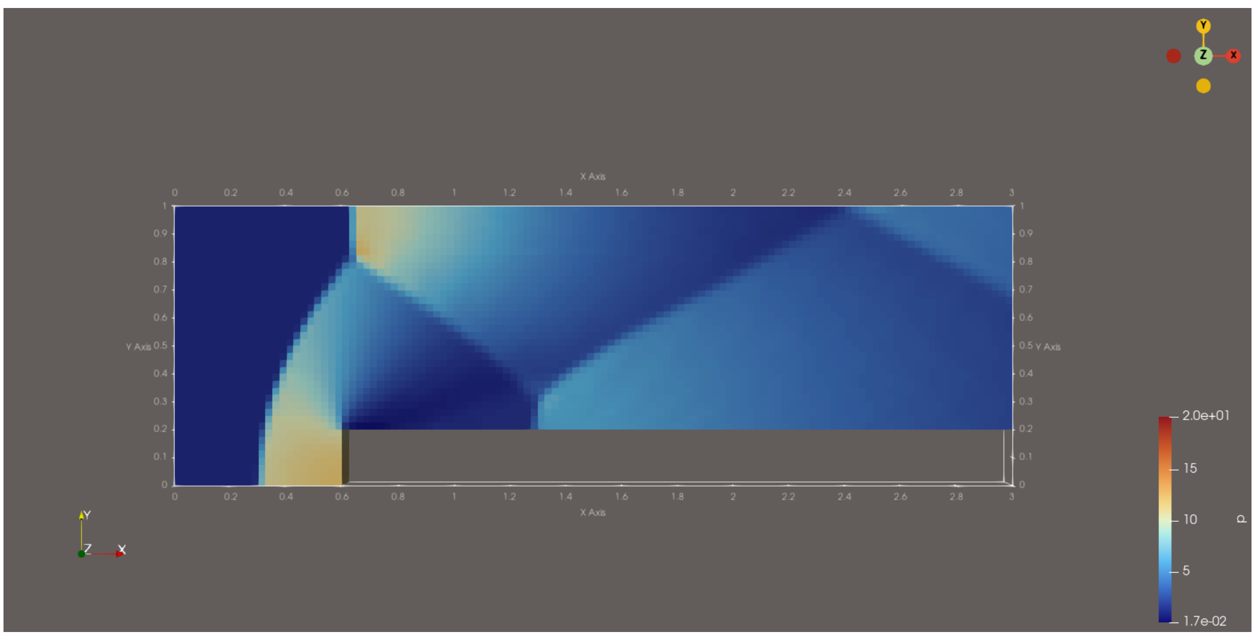

Figure 9: Pressure p [Pa] at $T = 4$ s — coarse mesh.

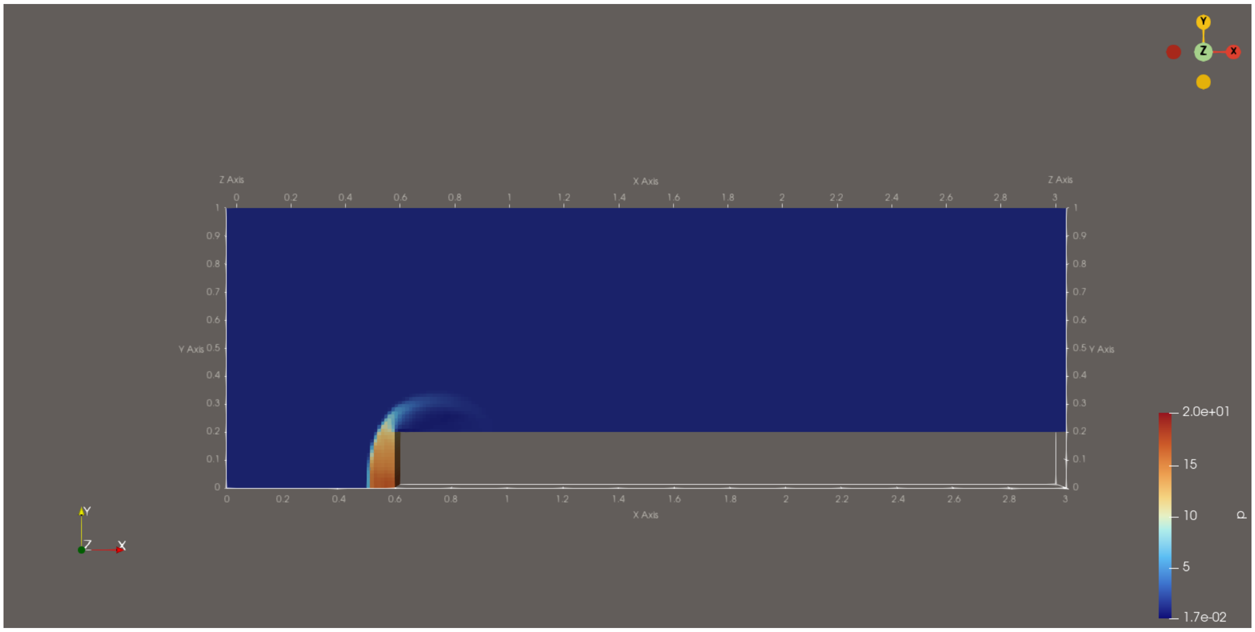

Figure 10: Pressure p [Pa] at $T = 4$ s — medium mesh.

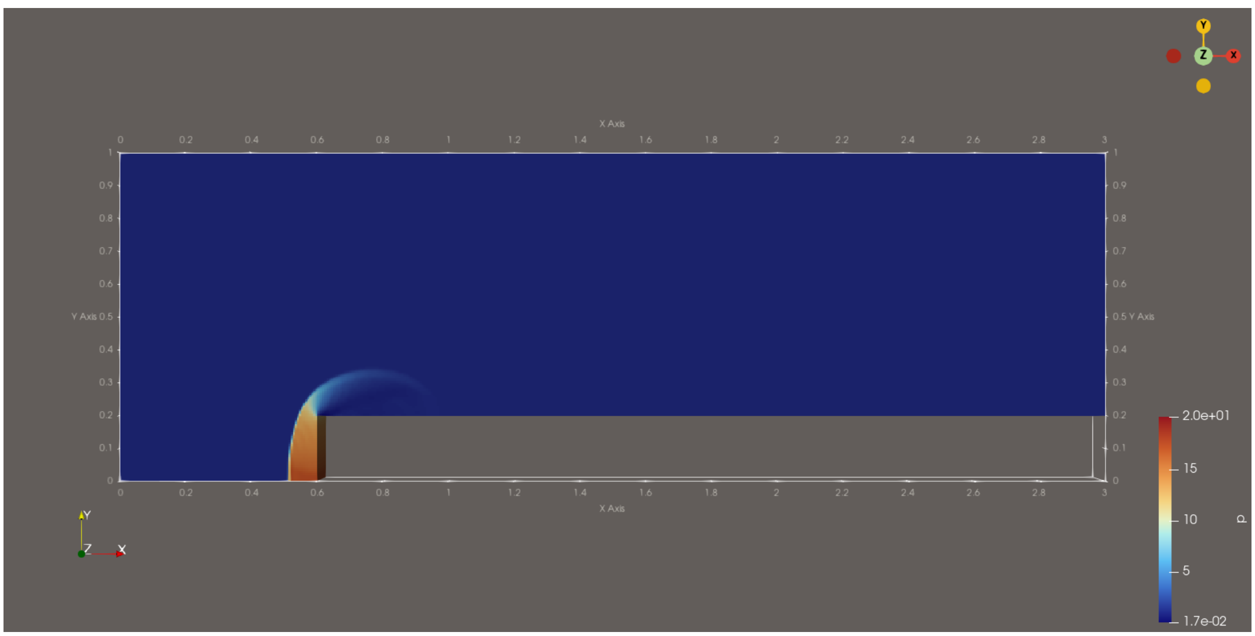

Figure 11: Pressure p [Pa] at $T = 4$ s — fine mesh.

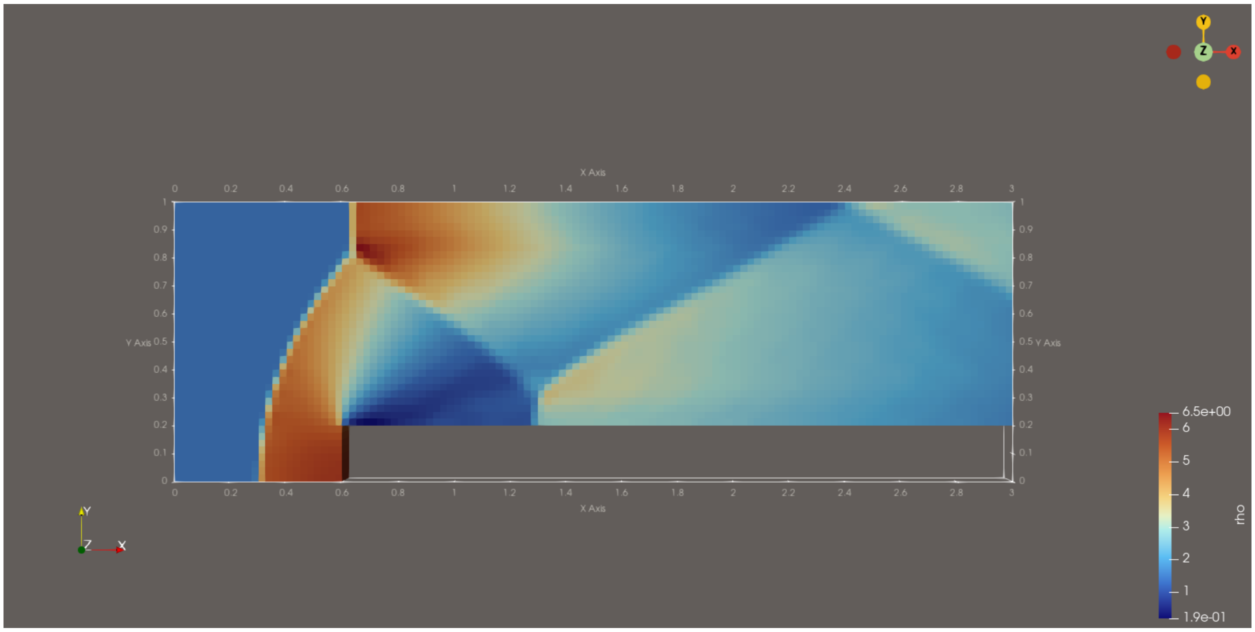

Figure 12: Density ρ [kg/m^3] at $T = 4$ s — coarse mesh.

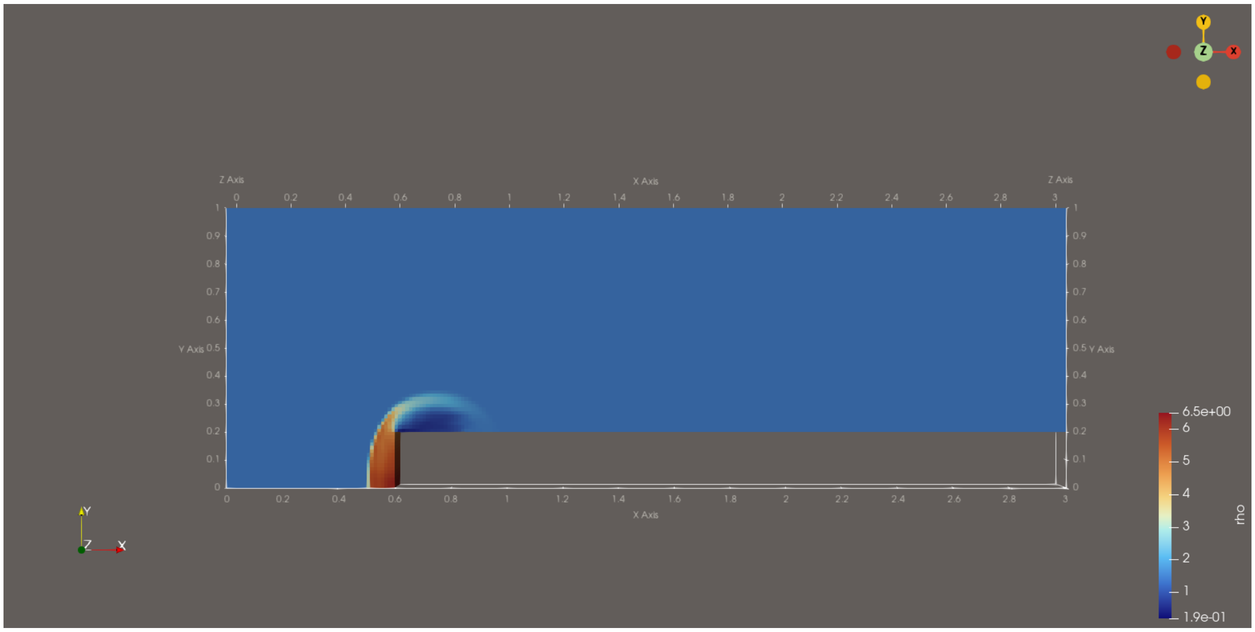

Figure 13: Density ρ [kg/m^3] at $T = 4$ s — medium mesh.

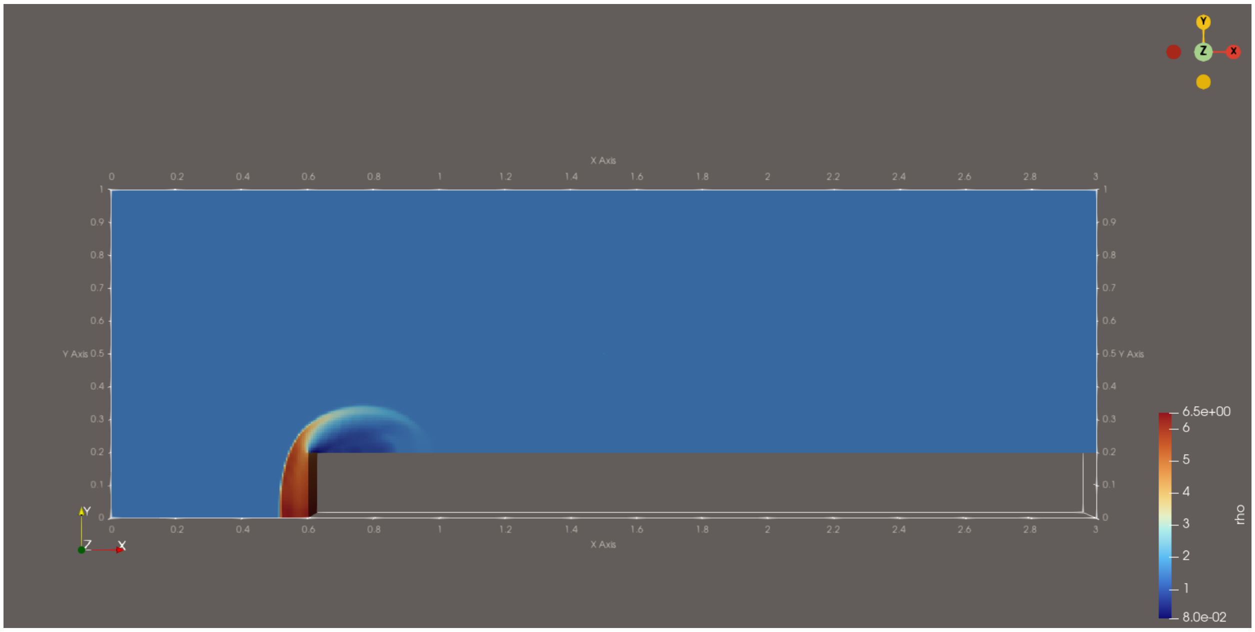

Figure 14: Density ρ [kg/m^3] at $T = 4$ s — fine mesh.

The effect of spatial resolution on the flow field is observed by comparing pressure and density contours across the coarse, medium, and fine meshes at $T = 4$ s. As the mesh is refined, shock structures become sharper and more well-defined, particularly near the step and in the downstream region. The coarse mesh captures the general structure of the flow but exhibits more numerical diffusion, while the medium and fine meshes show improved resolution of discontinuities. The fine mesh produces the most physically accurate representation of shock interactions and expansion regions.

6.3 Pressure Distribution Along Wall Surfaces

The pressure at $T = 4$ s was sampled along all wall surfaces using `singleGraph` cuts through the first cell adjacent to each wall. Results for all mesh levels are overlaid to demonstrate convergence.

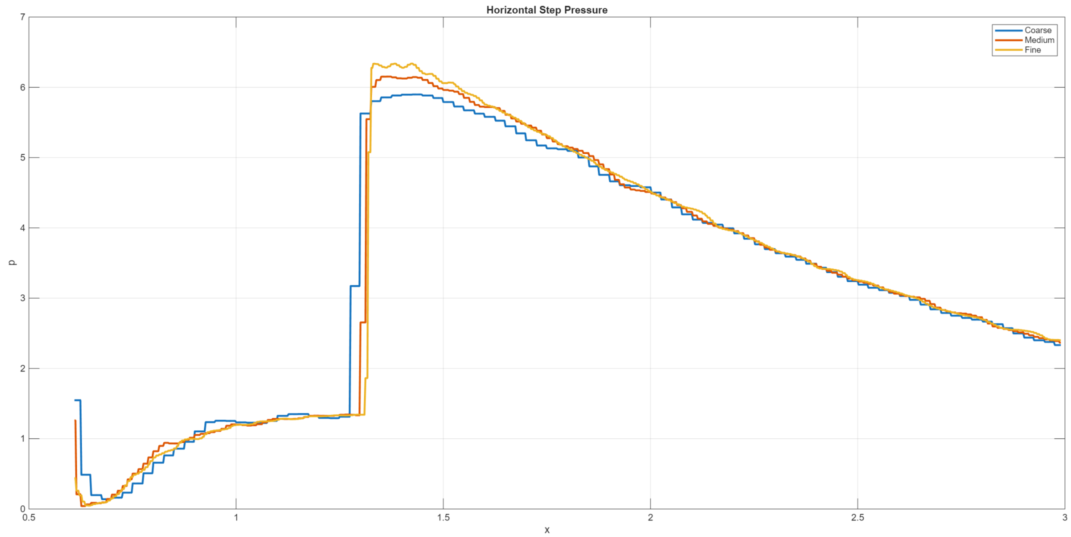

Figure 15: Pressure along the horizontal step face — coarse, medium, and fine meshes at $T = 4$ s.

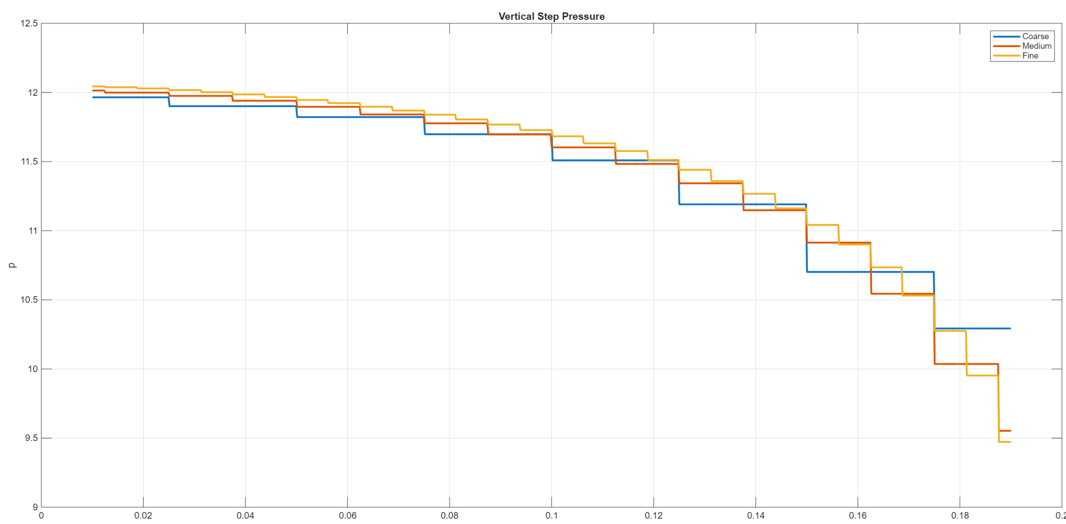

Figure 16: Pressure along the vertical step face — coarse, medium, and fine meshes at $T = 4$ s.

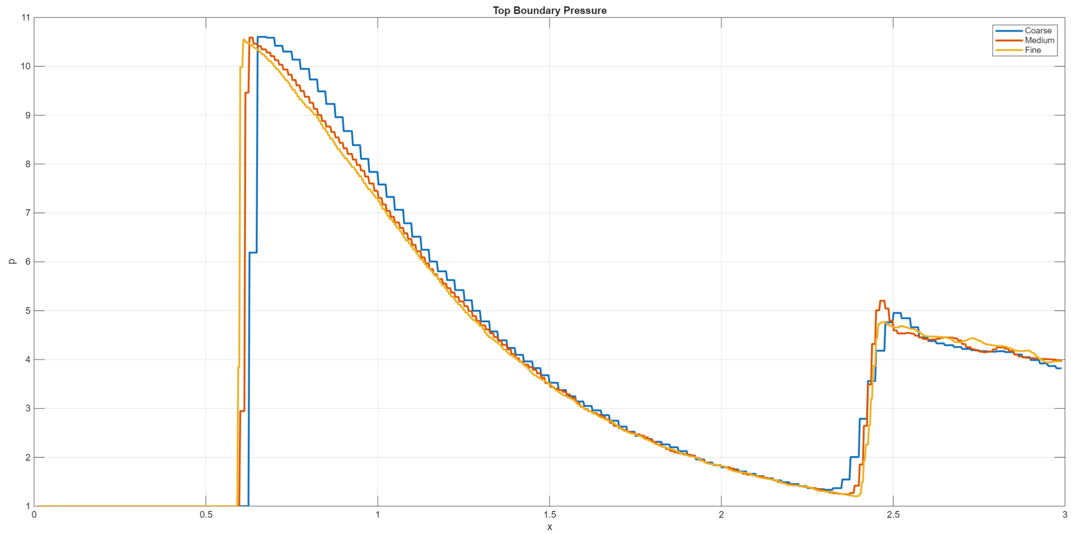

Figure 17: Pressure along the top boundary — coarse, medium, and fine meshes at $T = 4$ s.

The pressure at $T = 4$ s was extracted along the horizontal face of the step, the vertical face of the step, and the top boundary for all mesh levels. The results are plotted together to assess convergence.

The horizontal step surface exhibits a sharp pressure increase near the leading edge of the step due to shock formation and strong compression effects. The vertical face shows elevated pressure values associated with flow impingement and stagnation behavior. In contrast, the pressure along the top boundary remains nearly constant and close to the freestream value, which is consistent with the symmetry boundary condition and the reduced influence of shock interactions in that region.

The pressure distributions for the coarse, medium, and fine meshes are nearly indistinguishable, indicating that the solution has effectively converged with respect to spatial resolution. This suggests that further mesh refinement would not significantly change the wall pressure predictions.

6.4 Comparison with Analytical Stagnation Pressure

The numerical stagnation pressure was extracted from the vertical face of the step ($x = 0.6$ m) in the fine-mesh simulation at $T = 4$ s. The maximum pressure on this face, which occurs at the base of the step near $y = 0$ where the flow stagnates, is:

$$p_{0,\,\text{numerical}} = 12.0440 \text{ Pa}$$

The analytically predicted stagnation pressure (Section 4.3), derived from normal shock and isentropic flow relations for $M_1 = 3$ and $p_1 = 1$ Pa, is:

$$p_{0,\,\text{analytical}} = 12.0610 \text{ Pa}$$

The percent difference between the two values is **0.14%**, indicating excellent agreement between the simulation and theory. This small discrepancy is consistent with the finite spatial resolution of the numerical mesh and the slight displacement of the sampling line from the exact stagnation point at $y = 0$.

7 Parallel Solver Performance

The parallel performance of `rhoCentralFoam` was evaluated on a mesh of approximately 64,000 cells. The time per time step $T(P)$ was extracted from `log.rhoCentralFoam` for each processor count $P \in \{1, 2, 4, 8, 16, 32, 56\}$.

Parallel speedup and efficiency are defined as:

$$S(P) = \frac{T(1)}{T(P)}, \qquad \eta = \frac{S(P)}{P} = \frac{T(1)}{P\,T(P)} \tag{13}$$

To evaluate the scalability of the `rhoCentralFoam` solver, we conducted a series of parallel performance runs on TACC Frontera. The simulation was run from $t = 0$ to 0.3 s on the fine mesh (64,512 cells) with $Co_{\max} = 0.2$, producing 1008 adaptive time steps for each run. Processor counts of $P = 1, 2, 4, 8, 16, 32$, and 56 were tested. The domain was decomposed using hierarchical decomposition via OpenFOAM's `decomposePar` utility. For each run, the wall-clock time $T(P)$ was recorded using the `ClockTime` reported by OpenFOAM, and the speedup was computed as $S(P) = T(1)/T(P)$ with efficiency $\eta = S(P)/P$.

Table 3 summarizes the results for each processor count P.

Table 3: Parallel performance of `rhoCentralFoam` (≈64,000 cells).

P	**Total Clocktime (s)**	**# Steps**	$T(P)$ **(s/step)**	**Speedup** $S(P)$	**Efficiency** η
1	72	1008	0.07143	1.000	1.000
2	38	1008	0.03770	1.895	0.947
4	21	1008	0.02083	3.429	0.857
8	11	1008	0.01091	6.545	0.818
16	8	1008	0.00794	9.000	0.563
32	5	1008	0.00496	14.400	0.450
56	13	1008	0.01290	5.538	0.099

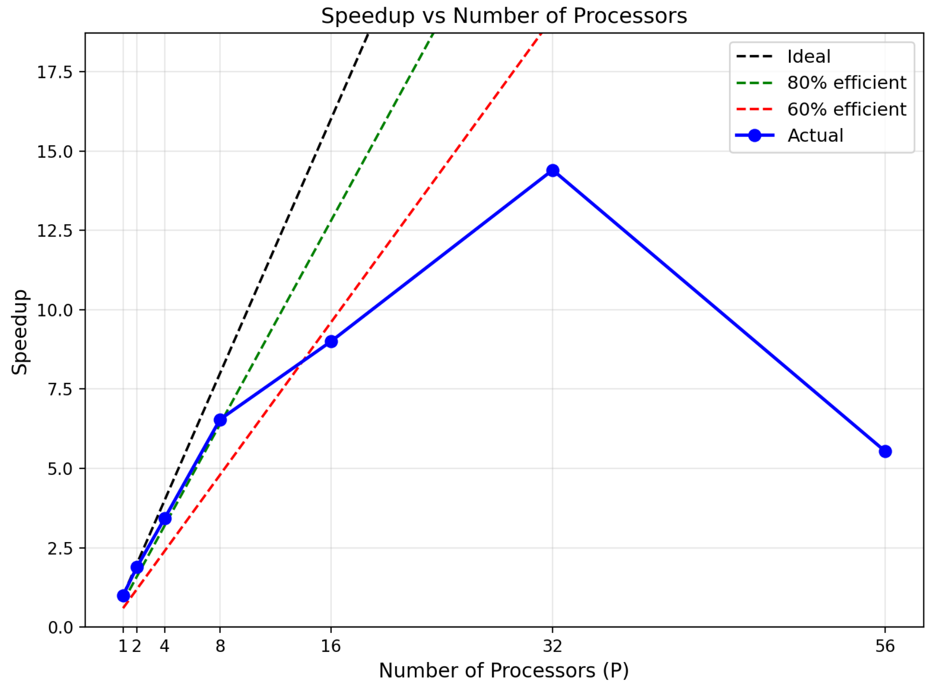

Figure 18: Speedup $S(P) = T(1)/T(P)$ vs. number of processors P — actual, ideal, 80% efficient, 60% efficient.

The solver demonstrates good parallel performance for low-to-moderate processor counts. From $P = 1$ to $P = 8$, efficiency remains above 80%, indicating that the computational workload per processor significantly outweighs the MPI communication overhead in this range. The speedup reaches 6.5× at $P = 8$ and continues to increase to a peak of 14.4× at $P = 32$.

Beyond $P = 8$, efficiency declines more rapidly. This is expected behavior: as the number of subdomains increases, each processor is responsible for a smaller portion of the mesh, while the number of inter-processor boundary faces — and thus the volume of MPI communication — grows. At $P = 16$ the efficiency is 56.3%, and at $P = 32$ it drops to 45.0%, reflecting this increasing communication burden.

At $P = 56$ (full node utilization), performance degrades significantly: the wall-clock time increases to 13 s compared to 5 s at $P = 32$, and the speedup falls to 5.5× with an efficiency of under 10%. This degradation is attributable to the small mesh size. With 64,512 cells divided across 56 processors, each core handles only approximately 1,150 cells. At this ratio, the cost of ghost-cell data exchange across subdomain boundaries at every time step exceeds the computational work, causing the parallel overhead to dominate. This represents the communication-bound regime and sets a practical upper limit on useful parallelism for this mesh resolution.

These results highlight an important consideration in parallel CFD: the optimal number of processors depends not only on hardware but on problem size. For the 64,512-cell mesh used here, $P = 16$ to $P = 32$ represents the best balance between speedup and efficiency. Scaling to a full 56-core node would require a significantly larger mesh to remain in the computation-bound regime.